

EE4410 CYBER SECURITY OF POWER GRIDS

Assignment 3: Cyber-Physical Data Correlation

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[Link to Google Colab Notebook](#)

This lab report looks at how cyber and physical signals in power grids relate to each other using data analysis and modeling.

1 Familiarization and data visualization

In the first part of this lab, the CPS dataset was loaded and analyzed to understand the main types of signals it contains and how they behave when things are working correctly versus when problems occur. The dataset consists of two types of signals:

- **Cyber signals:** represented by the `numpckt` variable, which records the number of GOOSE packets transmitted every 100 ms.
- **Physical signals:** represented by voltage values from buses 4 to 9, sampled during events. These continuous signals reflect the electrical behavior of the power system under operational or disturbed conditions.

When analyzing these signals, it becomes clear that there is both structure and correlation present, particularly under normal conditions. The GOOSE traffic during normal operation shows a regular, repeating burst pattern with values between zero and three packets per interval, indicating scheduled message exchanges between devices (see Figure 1).

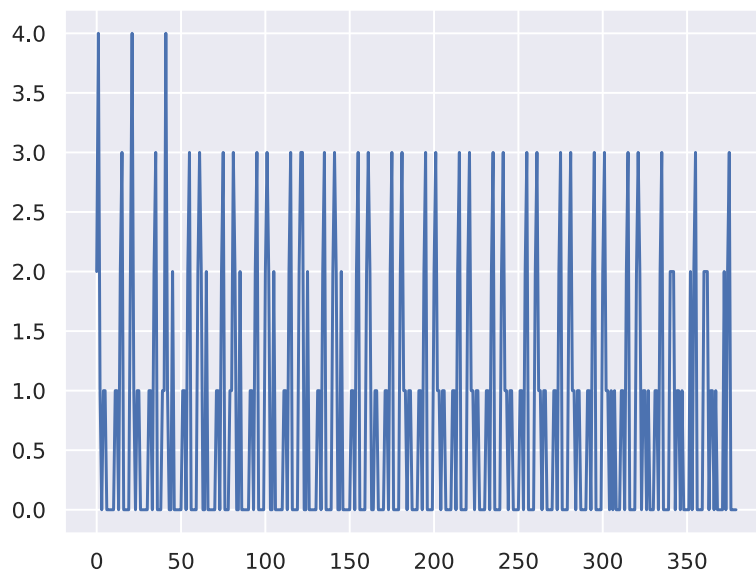
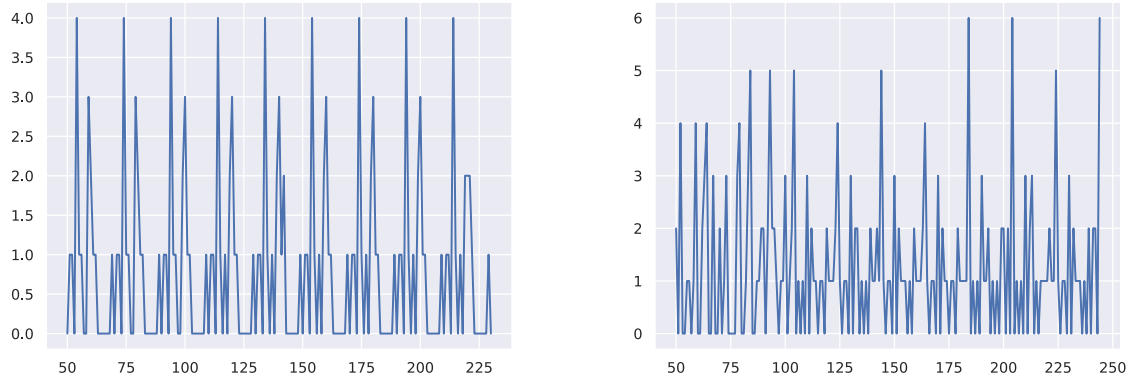


Figure 1: Normal GOOSE traffic

Event A traffic becomes slightly denser but still maintains a semi-periodic form, while Event B traffic is highly irregular and bursty, with peak values exceeding six packets per interval, as illustrated in Figure 2.



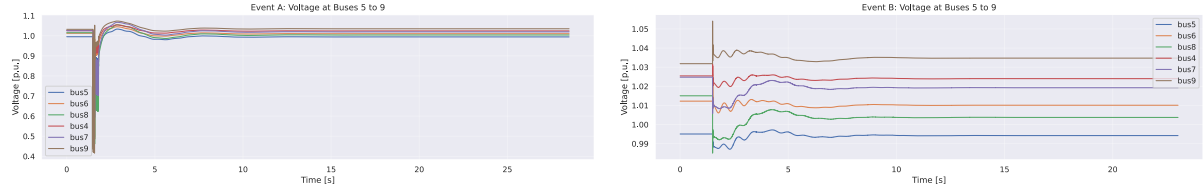
Event A GOOSE traffic

Event B GOOSE traffic

Figure 2: GOOSE traffics plots of Events A and B

This lack of structure in Event B traffic suggests a more chaotic or artificial communication pattern, which may be associated with spoofed or compromised messages.

The voltage plots are shown in Figure 3. During Event A, all monitored buses experience a sudden and dramatic voltage dip to around 0.4 p.u. at approximately $t = 2$ seconds, followed by a quick recovery and steady stabilization by $t = 6$ seconds. This is characteristic of a short-circuit fault that was quickly cleared by protection systems. Event B, by contrast, shows much milder voltage variations. While there is a clear disturbance, it does not involve a sharp drop but rather a series of low-amplitude oscillations. This pattern is consistent with a transient switching operation or an artificial event introduced in the system, possibly through the cyber layer.



Voltage at buses 5–9 during Event A

Voltage at buses 5–9 during Event B

Figure 3: Voltage magnitude plots across all buses during Events A and B

These signals suggest the datasets include both real-world equipment failures and cyberattacks. Event A probably represents a typical equipment failure, leading to a physical problem and a normal cyber reaction. On the other hand, Event B shows only a small physical problem but a very strong and unusual cyber reaction, which could indicate a spoofing attack or cyber interference.

2 ARMA Model

To identify and measure unusual patterns in GOOSE traffic under various conditions, an ARMA model was trained using data from normal operations. Before the training, the autocorrelation function (ACF) and partial autocorrelation function (PACF) were plotted in Figure 4 to select appropriate model parameters. Both ACF and PACF plots showed a sharp initial lag followed by rapid decay. Based on this, ($p=1$, $q=1$) were chosen.

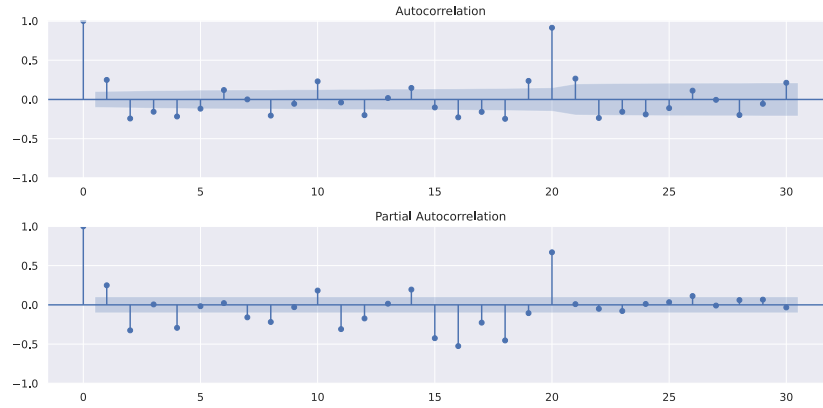


Figure 4: ACF and PACF plots

The model was trained on 80% of the normal data and predicted the remaining 20%. As shown in Figure 5, the ARMA predictions remained nearly constant and didn't reflect the bursty, irregular patterns of the real traffic. Although the errors were centered around zero, they were often significant, meaning the predictions weren't very accurate. This indicates that the model mainly learned the overall average and didn't properly capture the short-term changes in GOOSE traffic.

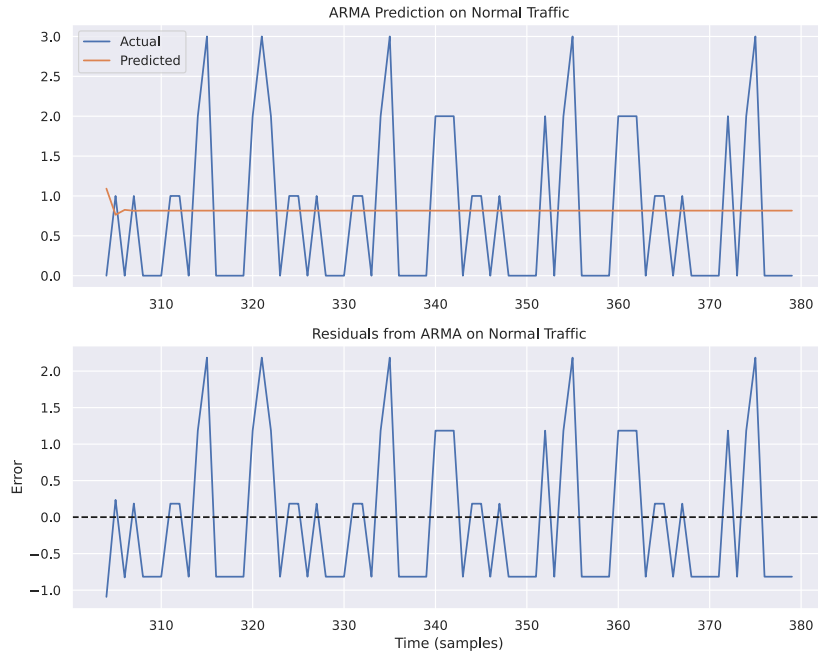


Figure 5: ARMA prediction and residuals on normal traffic

To test the model's ability to detect anomalies, it was applied to Event A and Event B data without retraining. In Event A, the residuals spiked significantly at the start of the event, but quickly decayed (Figure 6). This timing coincides with a voltage fault, showing that ARMA can identify brief disruptions.

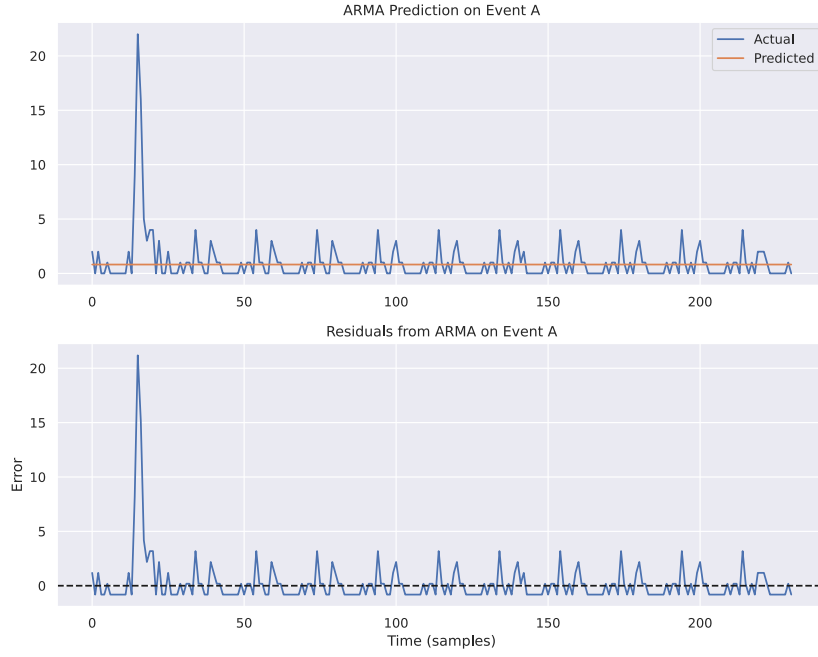


Figure 6: ARMA prediction and residuals for Event A. A sharp error spike reveals the presence of a fault

In contrast, Event B residuals were large and persisted throughout the signal (Figure 7), which indicates irregular and burst-like GOOSE activity that the model couldn't accurately represent. This suggests a possible cyber-induced anomaly, such as spoofing or injection.

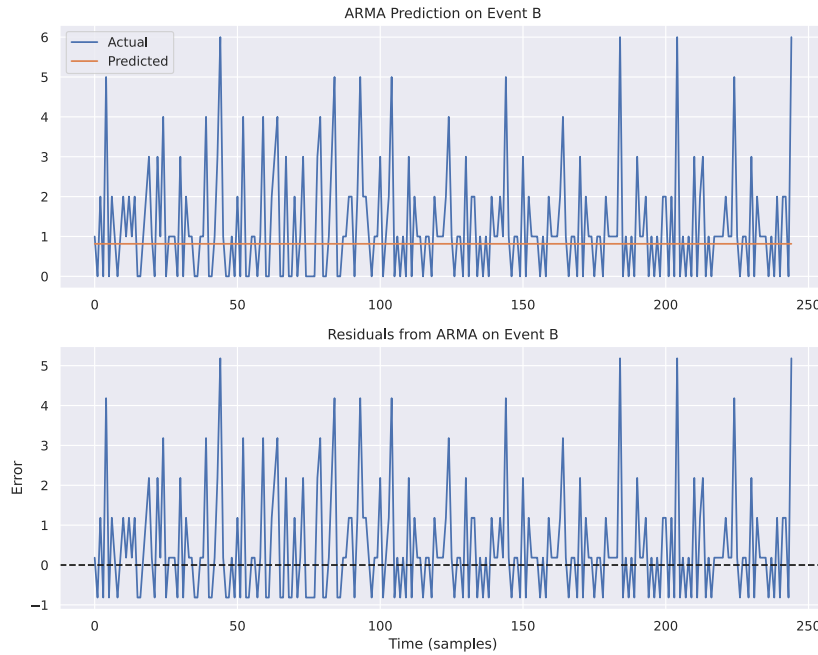


Figure 7: ARMA prediction and residuals for Event B. Persistent high errors indicate abnormal traffic

In summary, the $ARMA(1,1)$ model captured the general trend of normal traffic but did not accurately reproduce its short-term burstiness. However, it still proved useful for anomaly detection: brief residual spikes were visible during physical faults (Event A), and persistent high residuals revealed prolonged irregular activity in Event B. This indicates that even a simple model can highlight potential anomalies when the residual patterns are carefully analyzed.

3 SARMA Model

A Seasonal ARMA (SARMA) model was trained on the same normal traffic data using parameters ($p=1$, $d=0$, $q=2$) with seasonal order ($P=1$, $D=0$, $Q=2$, $s=4$) to explore potential improvements over ARMA.

While SARMA produced smaller residuals (ranging from -1 to $+2.5$) compared to ARMA's larger spikes, the model failed to accurately predict the actual traffic patterns (Figure 8). The predictions showed constant oscillations that ignored the irregular, event-driven nature of GOOSE traffic, essentially fitting a repetitive pattern rather than following the data.

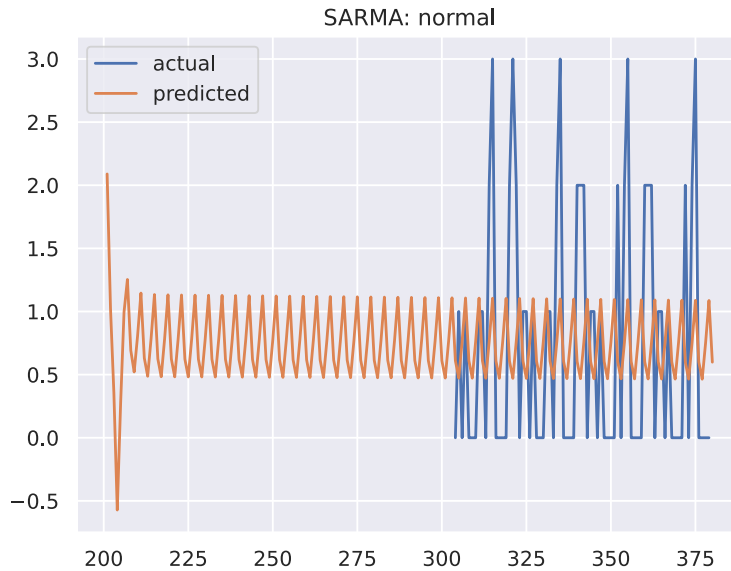


Figure 8: SARMA prediction on normal traffic

Although SARMA's residuals were numerically smaller and more stable (Figure 9), this came at the cost of predictive accuracy. The model's inability to track actual traffic variations makes it unsuitable for anomaly detection, as it would fail to produce the clear residual spikes needed to identify cyber attacks or equipment failures.

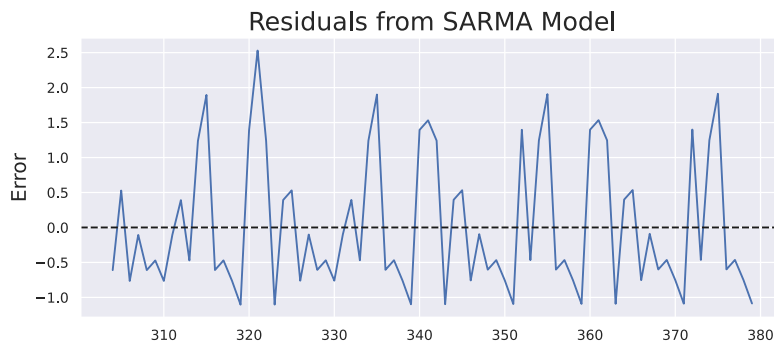


Figure 9: Residuals from SARMA model. Errors remain non-zero and irregular

The SARMA model produced smoother but less accurate predictions than ARMA. While SARMA's residuals were smaller in magnitude, they indicate a poorer fit to the bursty nature of GOOSE traffic because the model failed to track actual data variations.

SARMA models are generally more suitable when the data exhibits clear seasonal or periodic patterns, which is not the case here. Since GOOSE traffic is event-driven rather than seasonal, ARMA was better suited for modeling and detecting anomalies in this dataset.